

PAPO: A Policy-Aware Personalized Opportunity Framework for Heatwave Public Health Interventions

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Abstract

Extreme heat events are increasing in frequency and severity due to climate change, disproportionately affecting older adults and individuals with chronic conditions. While digital surveillance and AI systems can process environmental and health data in real time, public health policies and institutional response mechanisms often evolve slowly. This mismatch creates delays, fragmented guidance, and cognitive burden for both practitioners and citizens. We propose the Policy-Aware Personalized Opportunity (PAPO) framework, a middle-range, design-oriented AI framework that translates high-level public health policy into personalized, actionable intervention opportunities during heatwaves. PAPO emphasizes opportunity creation rather than risk signaling, aligning institutional intent with real-time systems and individual contexts. Using a heatwave case vignette, we illustrate how PAPO bridges the gap between slow institutional decision-making and fast-changing environmental and individual needs, while maintaining human oversight and institutional accountability.

Keywords: *PAPO framework, Policy-aware personalized opportunity, Middle-range theory, Heatwave public health interventions, AI-enabled decision support, Climate resilience, Vulnerable populations, Public health informatics*

1. Introduction

Heatwaves are among the most direct and deadly climate-related threats to public health. Older adults and individuals with chronic conditions face disproportionate risks, especially when mobility limitations, inadequate housing, or social isolation are present. Public health agencies have long recognized these vulnerabilities and developed emergency policies, guidance, and response protocols.

At the same time, AI-enabled digital systems have advanced rapidly. Environmental sensors, weather forecasting, and data integration platforms can now detect hazardous conditions in near real time. In theory, these capabilities enable timely and targeted public health interventions.

In practice, however, public health institutions face an institutional lag. Policies and guidelines are developed through deliberative processes and updated slowly, while digital systems operate continuously and at scale. Practitioners must interpret evolving conditions through fragmented alerts and delayed guidance, and citizens—particularly the most vulnerable—are often left confused about which actions are feasible or appropriate.

This gap raises a central question for AI-enabled public health: *how can institutions translate slow, high-level policy into fast, personalized, and actionable opportunities during time-critical events such as heatwaves?*

The Policy-Aware Personalized Opportunity (PAPO) framework is proposed as an institutional adaptation layer to address this challenge. PAPO does not replace policy or human judgment. Instead, it provides a structured mechanism for translating institutional intent into individualized, feasible actions in real time, aligning policy speed with system speed and human needs.

Conceptually, PAPO is grounded in *middle-range theory*^{1,2}, operating between abstract policy principles and narrow technical implementations. The framework builds on models introduced in my forthcoming Springer textbook on *AI in public health*³. It provides a design-oriented example of how middle-range frameworks can translate high-level public health policy into actionable, context-sensitive interventions during rapidly evolving events such as heatwaves.

2. Conceptual Framework

Heatwave response exposes a three-way bottleneck: policy processes are slow, AI-enabled systems are fast, and citizens experience confusion and cognitive burden. Traditional public health approaches emphasize risk communication through thresholds, warnings, and advisories. While essential, these approaches assume that individuals can independently convert risk information into action, an assumption that frequently fails for vulnerable populations.

PAPO reframes this challenge by shifting from risk signaling to opportunity creation. Its core logic follows a structured pathway: *policy is encoded into rules, rules are evaluated against risk and context, and feasible opportunities are generated that lead directly to intervention.*

The framework is policy-aware, ensuring alignment with institutional priorities and accountability; personalized, integrating individual vulnerability and daily context; and opportunity-based, translating guidance into concrete, actionable micro-interventions. By reducing cognitive burden and aligning institutions with real-time systems, PAPO functions as a middle-range framework that connects public health theory with operational decision-making.

3. PAPO Framework Architecture

The Policy-Aware Personalized Opportunity (PAPO) framework translates high-level public health policy into individualized, actionable interventions during heatwaves (See Figure 1). It consists of four interlinked components:

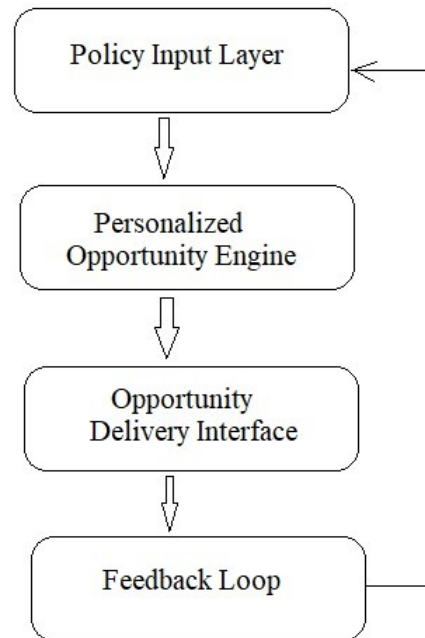


Figure 1. PAPO framework: four components translating policy into individualized, actionable opportunities during heatwave events.

1) Policy Input Layer

This layer encodes institutional intent, including protection priorities, eligibility criteria, and emergency thresholds. It integrates both static policy elements (e.g., cooling center rules and transportation protocols) and dynamic signals (e.g., real-time occupancy and grid alerts). Continuous alignment with policy ensures compliance and operational effectiveness.

2) Personalized Opportunity Engine

Combining individual vulnerability profiles (age, chronic conditions, mobility, routines) with real-time contextual signals (indoor temperature, location, symptom reports), this engine maps policy-constrained options into feasible, immediate micro-opportunities tailored to each individual.

3) Opportunity Delivery Interface

Opportunities are operationalized via multimodal channels—text, voice, and service referrals. Semi-autonomous, human-supervised workflows coordinate transport, community volunteer assistance, or wellness checks when standard alerts are insufficient, maintaining oversight and accountability.

4) Feedback Loop

Aggregated, privacy-preserving data on intervention uptake, response times, and resource use inform continuous policy refinement, resource allocation, and improved identification of

vulnerable populations. This adaptive loop ensures institutional guidance evolves with real-world conditions.

4. PAPO Application in a Heatwave Scenario

Extreme heat events evolve rapidly, while access to cooling resources is often uneven. Generalized alerts and public announcements frequently fail to reach those most at risk. The PAPO framework addresses this gap by aligning policy, system intelligence, and individualized action.

Consider Mr. Wu, an older adult living alone in a metropolitan area, with limited mobility, cardiovascular disease, and no functional air conditioning. Conventional heat alerts or cooling center notices are unlikely to prompt timely or feasible action. Using PAPO, institutional policy is translated into personalized, actionable opportunities, integrating Mr. Wu’s vulnerability profile with real-time environmental signals.

Table 1 below illustrates the operationalization of PAPO’s four components for this case:

PAPO Component	Inputs & Function	Actions & Outcomes
Policy Input Layer	Local heat emergency protocols, cooling center capacity, transportation policies. Encodes institutional intent and emergency rules.	Defines feasible interventions aligned with policy priorities. Ensure compliance.
Personalized Opportunity Engine	Mr. Wu’s age, chronic conditions, mobility, routines, preferred communication channels; real-time environmental data. Generate micro-opportunities.	Identifies actionable steps tailored to context. Recommendations are safe and timely.
Opportunity Delivery Interface	Outputs from engine; communication channels.	Delivers voice alerts, transport arrangements, volunteer notifications. Rapid coordinated interventions.
Feedback Loop	Intervention uptake, response time, resource utilization.	Aggregates anonymized data for system refinement. Improves responsiveness and equity.

Table 1. Operationalization of the PAPO framework during a heatwave: mapping each component to inputs, actions, and outcomes for an individual at high risk.

Using this approach, Mr. Wu receives immediate guidance and support. The system identifies a nearby staffed cooling center, arranges accessible transportation, and notifies a network of community volunteers under human supervision. The feedback loop captures anonymized data to refine future interventions, improving responsiveness for similarly vulnerable populations.

This scenario demonstrates PAPO’s role as a policy-aware, human-centered decision support framework. By converting high-level heat emergency policy into actionable, individualized

opportunities, PAPO bridges slow institutional processes and fast-changing environmental and individual needs, while maintaining accountability and transparency.

5. Governance, Safety, and Institutional Adaptation

PAPO is not autonomous or general intelligence. It is domain-specific, rule-based, and human-supervised. Transparency of decision rules, auditability, and safeguards against automation bias are core design requirements. As an institutional adaptation layer, PAPO helps public health systems align policy speed with AI system speed while preserving trust, accountability, and equity.

6. Implications for Climate Resilience and Digital Public Health

PAPO demonstrates how policy-aware, opportunity-oriented AI can enhance heatwave preparedness and climate resilience. By centering feasibility, personalization, and institutional governance, the framework offers a generalizable approach for time-critical public health challenges beyond heatwaves.

7. Limitations and Future Directions

This work presents a conceptual and design-oriented framework rather than a fully implemented system. As such, no empirical evaluation or real-world deployment results are reported. The primary contribution lies in articulating how policy-aware, personalized opportunity models can be structured to translate population-level heatwave policy into individual- and community-level decision support.

Future work will focus on maintaining an open, low-maintenance GitHub simulation repository providing CMDDS-PAPO-Heatwave (Conceptual Model Design Documentation for PAPO heatwave simulation). The repository is intended to support model-driven simulation and to encourage developers and researchers to fork the framework for locally contextualized heatwave scenarios, policy assumptions, and equity-aware risk analyses. Empirical evaluation, pilot deployment, and implementation studies—conducted through downstream forks or independent projects—will be required to assess effectiveness, equity impacts, and governance performance in applied settings.

8. Conclusion

PAPO reframes AI-enabled public health response from risk communication to opportunity creation. By translating policy into personalized, actionable interventions, the framework offers a practical path to protect vulnerable populations and support institutional adaptation in a fast-changing climate and AI era.

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